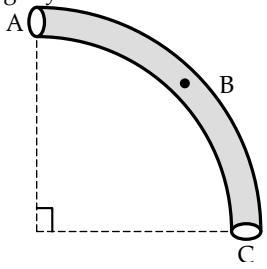
MOCK TEST PAPER 2

General Instructions: *Same as Mock Test Paper 1.*

Physics

Section A

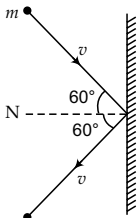
Q. 1. The tube AC forms a quarter circle in a vertical plane. Ball B has a cross-sectional area slightly smaller than that of the tube and can move through it without friction. B is placed at A and displaced slightly. It will:



(1) always be in contact with the inner wall of the tube
(2) always be in contact with the outer wall of the tube
(3) initially be in contact with the inner wall and later with the outer wall of the tube
(4) initially be in contact with the outer wall and later with the inner wall of the tube

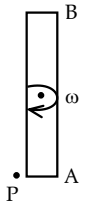
Q. 2. A galvanometer of 50 ohms resistance has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one division. To convert this galvanometer into a voltmeter with a range of 25 volts, it should be connected with a resistance of:
(1) 2500 Ω as a shunt **(2)** 2450 Ω as a shunt
(3) 2550 Ω in series **(4)** 2450 Ω in series

Q. 3. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed, as shown in the figure below. The value of impulse imparted by the wall on the ball will be:



(1) $\frac{mv}{2}$ **(2)** $\frac{mv}{3}$ **(3)** mv **(4)** $2mv$

Q. 4. A rod of mass 'M' and length 'L', lying on a frictionless horizontal surface, is initially given an angular velocity 'ω' about the vertical axis with it's centre of mass at rest but it's circular motion is not fixed. Subsequently, end A of the rod collides with nail P, which is near A, in such a way that end A becomes stationary immediately after impact. The velocity of end B just after collision will be:



(1) ωL **(2)** $\frac{\omega L}{2}$ **(3)** $\frac{\omega L}{4}$ **(4)** $\frac{7\omega L}{3}$

Q. 5. If R is the radius of the earth and g the acceleration due to gravity on the earth's surface, the mean density of the earth is:
(1) $\frac{4\pi G}{3gR}$ **(2)** $\frac{3\pi R}{4gG}$ **(3)** $\frac{3g}{4\pi RG}$ **(4)** $\frac{\pi Rg}{12G}$

Q. 6. A particle is oscillating according to the equation $x = 7 \cos 0.5\pi t$, where t is in seconds. The point moves from the position of equilibrium to maximum displacement in time:
(1) 4.0 second **(2)** 2.0 second
(3) 1.0 second **(4)** 0.5 second

Q. 7. A ball falling in a lake of 200 m shows a decrease of 0.1 % in its volume at the base of the lake. The bulk modulus of elasticity of the material of the ball is (take $g = 10$ m/s^2):
(1) 10^9 N/m^2 **(2)** 2×10^9 N/m^2
(3) 3×10^9 N/m^2 **(4)** 4×10^9 N/m^2

Q. 8. In a capillary tube, water rises to a height of 4 cm. If the cross-sectional area of the tube were one-fourth, water would have risen to a height of:
(1) 2 cm **(2)** 4 cm **(3)** 8 cm **(4)** 16 cm

Q. 9. Given below are two statements:

Statement I: The brightness of a bulb connected in series with a coil decreases when an iron rod is inserted into the coil.

Statement II: Inserting an iron rod into a coil increases the circuit's inductance, reducing the current flow and subsequently dimming the bulb.

Choose the correct answer from the options given below:

(1) Statement I is true, Statement II is true and Statement II is the correct explanation of Statement I.

(2) Statement I is true, Statement II is true, but Statement II is not the correct explanation of Statement I.

(3) Statement I is true, Statement II is false.

(4) Statement I is false, Statement II is true.

Q. 10. A solid cone of height 25 cm and base diameter 25 cm floats in water with its vertex downwards such that 20 cm of its axis is immersed. The additional weight that must be placed on the centre of the base such that the cone now is completely immersed in water is:

(1) 1 kg **(2)** 2 kg

(3) 3 kg **(4)** 4 kg

Q. 11. The electric field in a certain region is given by $\vec{E} = (5\hat{i} - 3\hat{j})$ kV/m. The potential difference $V_B - V_A$ between points A and B, having coordinates (4, 0, 3)m and (10, 3, 0)m respectively, is equal to

(1) 21 kV **(2)** –21 kV **(3)** 39 kV **(4)** –39 kV

Q. 12. Two monoatomic ideal gases 1 and 2 of molecular masses M_1 and M_2 are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is:

(1) $\sqrt{\frac{M_1}{M_2}}$ **(2)** $\sqrt{\frac{M_2}{M_1}}$ **(3)** $\frac{M_1}{M_2}$ **(4)** $\frac{M_2}{M_1}$

Q. 13. The electric field of an electromagnetic wave in free space is represented as $E_0\cos(\omega t - kz)\hat{i}$. The corresponding magnetic induction vector will be:

(1) $\vec{B} = E_0 c\cos(\omega t - kz)\hat{j}$

(2) $\vec{B} = \frac{E_0}{c}\cos(\omega t - kz)\hat{j}$

(3) $\vec{B} = E_0 c\cos(\omega t + kz)\hat{j}$

(4) $\vec{B} = \frac{E_0}{c}\cos(\omega t + kz)\hat{j}$

Q. 14. The wavefront of a light beam is given by the equation $x + 2y + 3z = C$ (where C is arbitrary constant), then the angle made by the direction of light with the y-axis is:

(1) $\cos^{-1}\frac{1}{\sqrt{14}}$ **(2)** $\sin^{-1}\frac{2}{\sqrt{14}}$

(3) $\cos^{-1}\frac{2}{\sqrt{14}}$ **(4)** $\sin^{-1}\frac{3}{\sqrt{14}}$

Q. 15. A film projector magnifies a 100 cm^2 film strip on a screen in such a way that the distance between the screen and the projector is divided in the ratio of 2:1 by the lens. Then the area of magnified film on the screen is:

(1) 1600 cm^2 **(2)** 400 cm^2

(3) 800 cm^2 **(4)** 200 cm^2

Q. 16. What is the number of wavelengths that can be emitted by hydrogen atoms when an electron falls from the fifth orbit to its ground state?

(1) 4 **(2)** 5 **(3)** 10 **(4)** 3

Q. 17. If the short series limit of the Balmer series for hydrogen is 3646 Å, calculate the atomic number of the element which gives X-ray wavelength down to 1.0 Å. Identify the element:

(1) $Z = 21$ **(2)** $Z = 31$

(3) $Z = 11$ **(4)** $Z = 5$

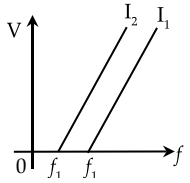
Q. 18. The wavelength of a neutron with energy 1 eV is closest to:

(1) 10^{-2} cm **(2)** 10^{-4} cm

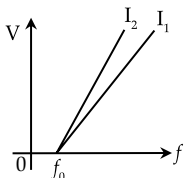
(3) 10^{-6} cm **(4)** 10^{-8} cm

Q. 19. A photoelectric experiment is performed at two different light intensities I_1 and I_2 ($>I_1$). Choose the correct graph showing the variation of stopping potential versus frequency of light.

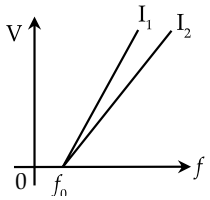
(1)



(2)



(3)



(4) None of these

Q. 20. If a semiconductor with an intrinsic carrier concentration of 1.41×10^{16} m^{-3} is doped with 10^{21} m^{-3} phosphorus, then the concentration of holes at room temperature will be:

(1) 2×10^{21} **(2)** 2×10^{11}

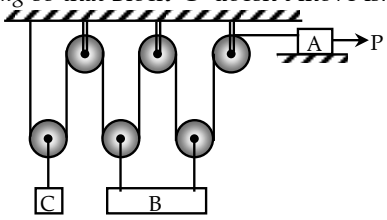
(3) 1.41×10^{10} **(4)** 1.41×10^{16}

Section B

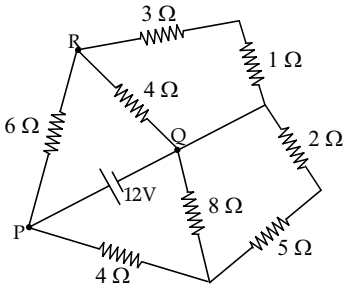
Q. 21. A ray of light passes through an equilateral prism (μ=1.5) such that $i = e$ and $e = \frac{3}{4}$ of angle of the prism. The angle of deviation is:

Q. 22. An Ideal Gas at 27°C is compressed adiabatically to $\frac{8}{27}$ of its original volume. The rise in temperature is (take γ =5/3):

Q. 23. Three blocks A, B and C, each with mass m, are arranged in a pulley mass system as shown. The coefficient of friction between Block A and the horizontal surface is equal to 0.5 and a Force P acts on 'A' in the direction shown. The value of P/mg so that Block 'C' doesn't move is.....



Q. 24. Power dissipated by the circuit is ..... W.



Q. 25. There is a constant homogeneous electric field of 100 Vm^{-1} within the region $x = 0$ and $x = 0.167$ m pointing in the positive x-direction. There is a constant homogeneous magnetic field B within the region $x = 0.167$ m and $x = 0.334$ m pointing in the z-direction. A proton at rest at the origin ($x = 0, y = 0$) is released in the positive x-direction. The minimum strength of the magnetic field B, so that the proton will come back to $x = 0$, $y = 0.167$ m (mass of the proton = 1.67×10^{-27} kg) is.....mT.

Chemistry

Section A

Q. 26. Hex–4–ene–2–ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B', which on further heating with soda lime gives 'C'. The compound 'C' is:

(1) 2–pentene **(2)** propionaldehyde
(3) 2–butene **(4)** 4–methylpent–2–ene

Q. 27. Which of the following pairs of species have the same bond order?

(1) N_2, NO^+ **(2)** O_2, NO^+
(3) N_2, O_2^- **(4)** CO, NO

Q. 28. The K_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in mol L^{-1} at 298 K is:

(1) 1.0×10^{-15} **(2)** 2.7×10^{-12}
(3) 3.2×10^{-10} **(4)** 4.2×10^{-8}

Q. 29. The element Z = 107 and Z = 109 have been made recently; element Z = 108 has not yet been made. Indicate the group in which you will place the above elements.

(1) 7, 8, 9 **(2)** 5, 6, 7
(3) 8, 9, 10 **(4)** 4, 5, 6

Q. 30. K_{a_1}, K_{a_2} and K_{a_3} are the respective ionisation constants for the following reactions (a), (b) and (c).

(a) $H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^-$

(b) $HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-}$

(c) $H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-}$

The relationship between K_{a_1}, K_{a_2}, and K_{a_3} is given as

(1) $K_{a_3} = K_{a_1} + K_{a_2}$ **(2)** $K_{a_3} = K_{a_1} - K_{a_2}$
(3) $K_{a_3} = K_{a_1} / K_{a_2}$ **(4)** $K_{a_3} = K_{a_1} \times K_{a_2}$

Q. 31. Enthalpy of fusion of a liquid is 1.435 kcal mol^{-1} and molar entropy change is 5.26 cal $mol^{-1}K^{-1}$. Hence melting point of liquid is:

(1) 100°C **(2)** 0°C
(3) 373°C **(4)** –273°C

Q. 32. In $Fe_4\left[Fe(CN)_6\right]_3$ the oxidation state of the complexed iron is:

(1) +3 **(2)** +2 **(3)** +4 **(4)** +6

Q. 33. Assertion A and the other is labelled as Reason R:

Assertion A: O_2^- is paramagnetic.

Reason R: O_2^- have all paired electrons.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is NOT the correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is the correct explanation of A.

Q. 34. Given below are two statements:

Statement I: The ionisation energy of Cu is more than that of K though both have a 4s configuration.

Statement II: The 18 electron shell of Cu shields nuclear charge more effectively than the shell of K.

Choose the correct answer from the options given below:

(1) Statement I is incorrect but statement II is true.

(2) Both statements I and II are true.

(3) Both statements I and statement II are false.

(4) Statement I is correct but statement II is false.

Q. 35. Ethyl methyl vinyl amine has the structure:

(1) $CH_3CH_2–N(CH_3)–CH_2CH=CH_2$

(2) $CH_3CH_2–N(CH_3)–CH=CH_2$

(3) $CH_2=CH–N(CH_3)–CH=CH_2$

(4) $CH_3–N(CH_3)–CH=CH_2$

Q. 36. The correct name of $(CO)_3Fe–CO–Fe(CO)_3$ is:

(1) Tri-μ-carbonyl bis (tricarbonyl iron (0))

(2) Hexacarbonyl iron (III) μ-tricarbonyl ferrate (0)

(3) Tricarbonyl iron (0) μ-tricarbonyl iron (0) tricarbonyl

(4) Nonacarbonyl iron

Q. 37. Given below are two statements:

Statement I: Complexes of type MA_5B and MA_6 do not show geometrical isomerism.

Statement II: Coordination compounds having coordination number 6 do not exhibit geometrical isomerism.

Choose the correct answer from the options given below:

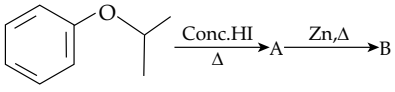
(1) Statement I is incorrect but statement II is true.

(2) Both statement I and statement II are true.

(3) Both statement I and statement II are false.

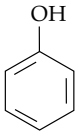
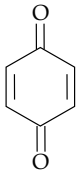
(4) Statement I is correct but statement II is false.

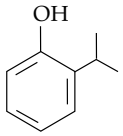
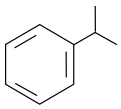
Q. 38. Compound I is heated with conc. HI to give a hydroxy compound A which is further heated with Zn dust to give compound B. Identify A and B.

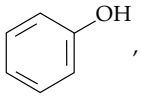


I

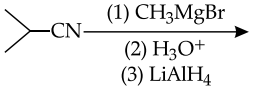
(1) A =  , B = 

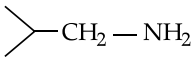
(2) A =  , B = 

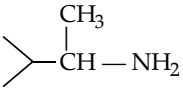
(3) A =  , B = 

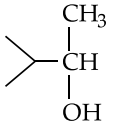
(4) A =  , B = 

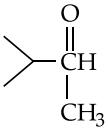
Q. 39. The major product of the following reaction is:



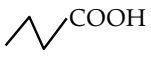
(1) 

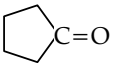
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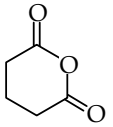
(3) 

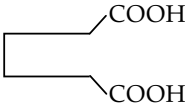
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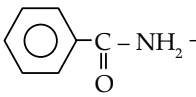
Q. 40. The product formed when adipic acid is heated:

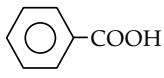
(1) 

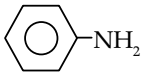
(2) 

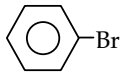
(3) 

(4) 

Q. 41.  product (A) is:

(1) 

(2) 

(3) 

(4) None the above

Q. 42. $(C_6H_{12}O_5)_n + nH_2O \xrightarrow{(X)} nC_{12}H_{22}O_{11}$

$C_{12}H_{22}O_{11} + H_2O \xrightarrow{(Y)} 2C_6H_{12}O_6$

Enzymes X and Y are:

(1) Zymase and Invertase

(2) Diatase and Maltase

(3) Maltase and Diatase

(4) Diatase and Zymase

Q. 43. Given below are two statements:

Statement I: At a frequency $v > v_0$, the ejected e^- come out with certain kinetic energy.

v = incident frequency; v_0 = threshold frequency

Statement II: The kinetic energy of these electrons increase with the increase of frequency of the light used.

(1) Statement I is correct and statement II is incorrect.

(2) Statement I is incorrect and statement II is correct.

(3) Both statement I and II are incorrect.

(4) Both statement I and II are correct.

Q. 44. In the Hunsdiecker reaction:

(1) number of carbon atoms decrease

(2) number of carbon atoms increase

(3) number of carbon atoms remain same

(4) none of the above

Q. 45. In the reaction sequence,

$$CaC_2 \xrightarrow{H_2O} A \xrightarrow[Hg^{+2}]{dil.H_2SO_4} B \xrightarrow[Ni]{H_2} C,$$

the product C is:

(1) CH_3OH **(2)** CH_3CHO

(3) C_2H_5OH **(4)** C_2H_4

Section B

Q. 46. Given that the temperature coefficient for the saponification of ethyl acetate by NaOH is 1.75. The activation energy for the saponification of ethyl acetate is ..... kcal mol^{-1}.

Q. 47. Cyclohexane-1,4-dione is a polar compound, having dipole moment value of 1.2 D. If mol fraction of its chair form is 0.80, the dipole moment of twisted boat form will be .....

Q. 48. The resistance of a conductivity cell containing 0.01 M KCl solution at 298 K is 1750Ω. If the conductivity of 0.01 M KCl solution at 298 K is 0.152×10^{-3} S cm^{-1}, then the cell constant of the conductivity cell is _____ $\times 10^{-3}$ cm^{-1}.

Q. 49. An organic liquid, A, is immiscible with water. When boiled together with water, the boiling point is 90°C at which the partial vapour pressure of water is 526 mm Hg. The atmospheric pressure is 736 mm Hg. The weight ratio of the liquid and water collected is 2.5: 1. The molecular weight of the liquid is ..... g.

Q. 50. The quantity of benzene, when 91.2 g of Phenylmagnesium iodide is treated with 4.2 g of Pent-4-yn-1-ol at STP would be produced ..... *l*.

Mathematics

Section A

Q. 51. The remainder when $32^{32^{32}}$ is divided by 7 is m. The value of limit $\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}}$ is e^n. The quadratic equation whose roots are $\frac{1}{m}$ and $\frac{1}{n}$ is:

(1) $x^2 - 3x - 4 = 0$ **(2)** $4x^2 + 3x - 1 = 0$

(3) $4x^2 - 3x + 1 = 0$ **(4)** $x^2 + 3x - 4 = 0$

Q. 52. $\int_0^1 \frac{dx}{(x^2-2x+2)^3} =$

(1) $\frac{3\pi+8}{32}$ **(2)** $\frac{\pi+1}{4}$

(3) 0 **(4)** $\frac{2\pi}{3}$

Q. 53. Consider system of linear equations: $x + y + z = 6$; $x + 2y + 3z = 10$; $x + 2y + k_1z = k_2$. Then, which one of the following is correct:

(1) Unique solution for $k_1 \neq 2$

(2) No solution for $k_1 \neq 3$ and $k_2 = 10$

(3) Infinite no. of solution for $k_1 = 3$ and $k_2 = 10$

(4) None of the above

Q. 54. If the solution of the differential equation $\frac{dy}{dx}(x^2y^3 + xy) = 1$ is $x(y^m + n + c_1e^{-y^2/k}) = l$, then ${}^{m-n+k+l}C_m$ is equal to (where c is constant of integration):

(1) 10 **(2)** 5

(3) 12 **(4)** Not defined

Q. 55. A triangle has two of its sides along the axis, its third side touches the circle

$x^2 + y^2 - 2\lambda x - 2\lambda y + \lambda^2 = 0$.

The equation of locus of the circumcentre of the triangle is $m(x+y) - \lambda = \frac{nxy}{\lambda}$, then $(m + n)$ is equal to:

(1) 4 **(2)** 3 **(3)** 0 **(4)** 6

Q. 56. Let PQR be a triangle. Points A, B and C are taken on the sides PQ, QR and RP respectively, such that $\frac{PA}{PQ} = \frac{QB}{QR} = \frac{RC}{RP} = m$. The value of m for which area of the triangle formed by $\vec{PB}$, $\vec{QC}$ and $\vec{RA}$ is the least is:

(1) $\frac{1}{3}$ **(2)** $\frac{1}{4}$ **(3)** $\frac{1}{2}$ **(4)** $\frac{1}{6}$

Q. 57. The root of the equation

$$[x\ 1\ 2]\begin{bmatrix}0 & 1 & 1\\ 1 & 0 & 1\\ 1 & 1 & 0\end{bmatrix}\begin{bmatrix}x\\ -1\\ 1\end{bmatrix} = 0 \text{ is .....}$$

(1) $\frac{1}{3}$ **(2)** $-\frac{1}{3}$ **(3)** 0 **(4)** 1

Q. 58. Consider points $P\left(\sqrt{13}, 0\right)$ and $Q\left(2\sqrt{13}, 0\right)$ lying on x-axis. These points are rotated in an anticlockwise direction about the origin through an angle $\tan^{-1}\left(\frac{2}{3}\right)$. Let the new position of P and Q be P′ and Q′ respectively. If the foot of perpendicular from (4, – 2) to line P′Q′ is (α, β), then 13 (α + β) is equal to:
(1) 30 **(2)** 40 **(3)** 45 **(4)** 25

Q. 59. From a pack of well shuffled cards, one card is drawn randomly. A gambler bets that it is either a diamond or a king. The odds in favour of his winning the bet are:
(1) 9:4 **(2)** 4:9 **(3)** 5:7 **(4)** 9:7

Q. 60. Let $z_1 = 10 + 6i$, $z_2 = 4 + 6i$ and z is a complex number such that $\arg\left(\frac{z-z_1}{z-z_2}\right)=\frac{\pi}{2}$, then the value of $|z - 7 - 9i|$ is..... .
(1) 2 **(2)** 6 **(3)** $3\sqrt{2}$ **(4)** 8

Q. 61. PQR is a triangle where P = (2, 3, 5), Q = (– 1, 2, 2) and R = (α, 5, β). If the median through P is equally inclined to the axis, then ${}^nC_0 + 3{}^nC_1 + 5{}^nC_2 +$..... + $(2n + 1)\, C_n$ is equal to, where $n = \alpha + \beta$:
(1) 2^{15} **(2)** 2^{18} **(3)** 2^{20} **(4)** 2^{19}

Q. 62. If $g : R \to \left(0, \frac{\pi}{2}\right]$, then the value of k for which $g(x) = \cot^{-1}(x^2 + x + k)$ is an onto function, is given by:
(1) $\frac{1}{3}$ **(2)** $\frac{1}{4}$ **(3)** $\frac{1}{8}$ **(4)** $\frac{1}{2}$

Q. 63. If a, b, c are in AP; a, b, d are in GP; then a, $a - b$, $d - c$ are in
(1) AP **(2)** GP
(3) HP **(4)** AP and GP both

Q. 64. $\frac{1}{1!\cdot(n-1)!} + \frac{1}{3!\cdot(n-3)!} + \frac{1}{5!\cdot(n-5)!} + \ldots$ is equal to:
(1) $\frac{2^{n-1}}{n!}$ for even values of n only
(2) $\frac{2^{n-1}+1}{n!}$ –1 for odd values of n only
(3) $\frac{2^{n-1}}{n!}$ for all $n \in N$
(4) $\frac{2^n}{(n-1)!}$ for all $n \in N$

Q. 65. If $\int \frac{\cos^2 x + \sin 2x}{(2\cos x - \sin x)^2} dx = \frac{1}{f(x)} - \frac{m}{5}\log_e f(x) + \frac{n}{5}\log_e (g(x)) - \frac{r}{5}h(x) + c$; where c is constant of integration and $m > n$, then which of the following is correct?
(1) $f(0) + g(0) + h(0) = 4$
(2) $m + n + r = 3$
(3) Minimum value of $g(x) = -2$
(4) $f'(0) + g'(0) + h'(0) = 0$

Q. 66. Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11} and B = {1, 3, 4, 5, 6}. Then, the number of subsets of B^C, which must contain 10, is:
(1) 16 **(2)** 32 **(3)** 64 **(4)** 8

Q. 67. The equation of the parabola whose focus is (1, 1) and tangent at the vertex is $x + y = 1$, is
(1) $x^2 + y^2 - 2xy - 4x - 4y + 4 = 0$
(2) $x^2 + y^2 - 2xy + 4x + 4y + 4 = 0$
(3) $x^2 + y^2 - 2xy - 4x - 4y - 4 = 0$
(4) $x^2 + y^2 - 2xy + 4x + 4y - 4 = 0$

Q. 68. The domain of the function $f(x) = \frac{1}{\sqrt{x - 2\{x\}}} + \ln\{x\}$ (where { } denotes fractional part function) is
(1) R – I **(2)** $(1, \infty)$
(3) $[1, \infty) - I^+$ **(4)** $(0, \infty)$

Q. 69. If the hyperbolas, $x^2 + 3xy + 2y^2 + 2x + 3y + 2 = 0$ and $x^2 + 3xy + 2y^2 + 2x + 3y + c = 0$ are conjugate of each other, then the value of 'c' is equal to:
(1) – 2 **(2)** 4 **(3)** 0 **(4)** 1

Q. 70. Five numbers x_1, x_2, x_3, x_4, x_5 are randomly selected from the numbers 1, 2, 3,..., 18 and are arranged in the increasing order ($x_1 < x_2 < x_3 < x_4 < x_5$). The probability that $x_2 = 7$ and $x_4 = 11$ is:
(1) $\frac{1}{136}$ **(2)** $\frac{1}{72}$ **(3)** $\frac{1}{68}$ **(4)** $\frac{1}{34}$

Section B

Q. 71. If $\sin^2(10°)\sin(20°)\sin(40°)\sin(50°)\sin(70°) = \alpha - \frac{1}{16}\sin(10°)$, then $16 + \alpha^{-1}$ is equal to _____.

Q. 72. If $\frac{f(x)}{\sin^2 x} = -\cos^{-1}\left(\frac{2\sqrt{2}x}{\pi}\right) - |f(x)|$, then the value of $\frac{f(\pi/4)}{(-\pi/32)}$ is ..... .

Q. 73. The value of $\lim\limits_{x\to 2}\frac{2^x + 2^{3-x} - 6}{\sqrt{2^{-x}} - 2^{1-x}}$ is ..... .

Q. 74. If complex number z_1 and z_2 satisfy the equations $|z_1| = 6$ and $|z_2 - 4| + |z_2 + 4| = 10$, then the maximum value of $|z_1 - z_2|$ is ..... .

Q. 75. If $\ln\left((e-1)e^{xy} + x^2\right) = x^2 + y^2$, then $\left(\frac{dy}{dx}\right)_{(1,0)}$ is equal to ..... .